

24-03-05

Sunday, June 14, 2026 8:59 AM

DC-DC converters

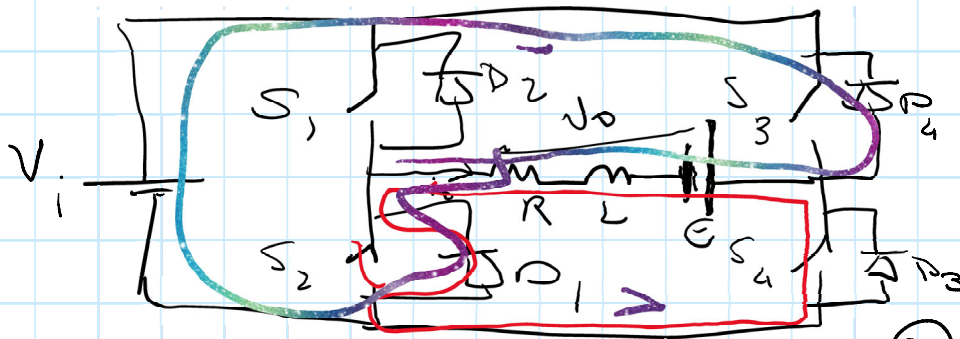
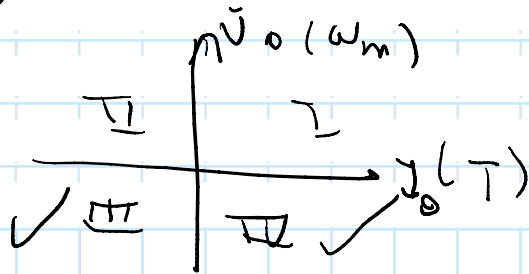
Choppers

power supplies

1Q

2Q

4Q



① in each mode one switch (main switch) & one diode operate alternate to the main switch.

③ in III, IV, polarity & changes

② in motoring mode (I, II) π, π

S on $\rightarrow V_o$ S on $\rightarrow \phi$

S off $\rightarrow \phi$ S off $\rightarrow -V_o$

Q1 operation:

S_1, D_1 main switches
 S_4 is always closed.

S_1 on $\rightarrow V_o = V_i$
 D_1 off

S_1 off $V_o = \phi$, $\bar{V}_o = D_{S1} V_i$
 D_1 on $M = D_{S1}$

Q2 operation

S_2, D_2 main switches

S_2 on D_3 always on

D_2 off $V_o = \phi$

S_2 off $V_o = -V_i$ $\bar{V}_o = (1 - D_{S2}) V_i$
 D_2 on $M = 1 - D_{S2}$

Q3 operation

S_3, D_3 main switches

S_3, S_2 on $V_o = -V_i$
 D_3 off

S_2, D_3 on $V_o = \phi$ $\bar{V}_o = D_{S3} V_i$
 S_3 off $M = -D_{S3}$

W3 2/8

2/3 2/8

$$m = -D_{S3}$$

Q4 operation:

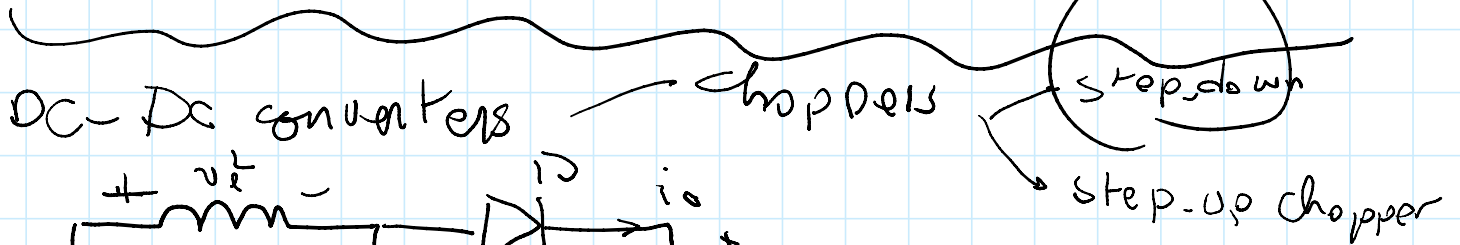
S_4, D_4 main switches

$$S_4, D_4 \text{ on} \rightarrow V_o = 0$$

$$|V_d| = (1 - D_{S4}) V_i$$

$$D_4, S_4 \text{ on} \rightarrow V_o = -V_i$$

$$m = -(1 - D_{S4}) V_i$$

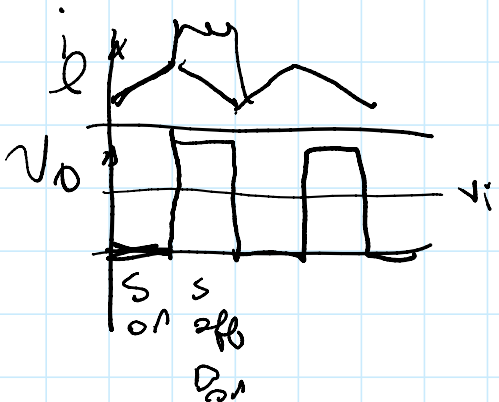


S on $\rightarrow L$ is charged $\rightarrow V_o = 0$

S off $\rightarrow L$ is discharged $\rightarrow V_o = -V_L + V_i$

$$L \frac{di_L}{dt} < 0$$

$$|V_o| = L \left| \frac{di_L}{dt} \right| + V_i$$



$$S \text{ on} \quad V_i = L \frac{di_L}{dt} \approx L \frac{\Delta i_L}{t_{on}}$$

$$\Delta i_L = \frac{V_i}{L} t_{on}$$

$$S \text{ off} \quad L \frac{di_L}{dt} \approx L \frac{\Delta i_L}{t_{off}} = V_i - V_o$$

$$\Delta i_L = \frac{V_i - V_o}{L} t_{off}$$

$$-x_2 = \frac{V_i}{L} T_{off}$$

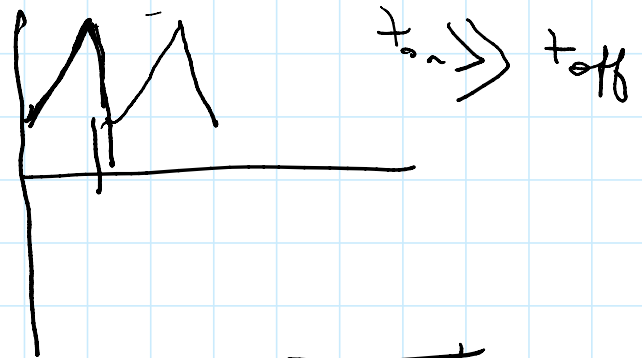
$| \Delta i_{L1} | = | \Delta i_{L2} |$ @ steady state

$$\frac{\Delta i_{L2}}{\Delta t} = - \frac{\Delta i_{L1}}{T_{off}} = - \frac{V_i}{L} \frac{T_{on}}{T_{off}}$$

$$V_o = V_i - V_d = V_i - L \frac{di_L}{dt} \Rightarrow V_o = V_i + L \frac{|\Delta i_{L2}|}{T_{off}}$$

$$V_o = V_i \left(1 + \frac{T_{on}}{T_{off}}\right) \Rightarrow \boxed{V_o = \frac{V_i}{1-D_s}}$$

$$D_s \rightarrow 1 : V_o \rightarrow \infty$$



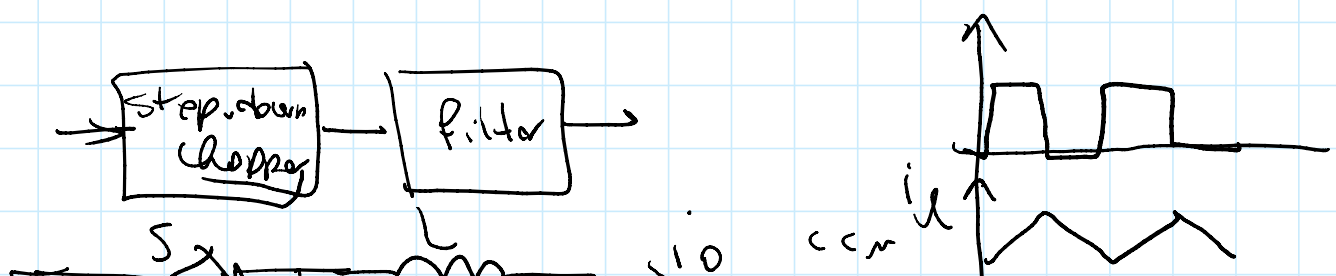
$$V_o = V_i + L \left| \frac{\Delta i_L}{T_{off}} \right|$$

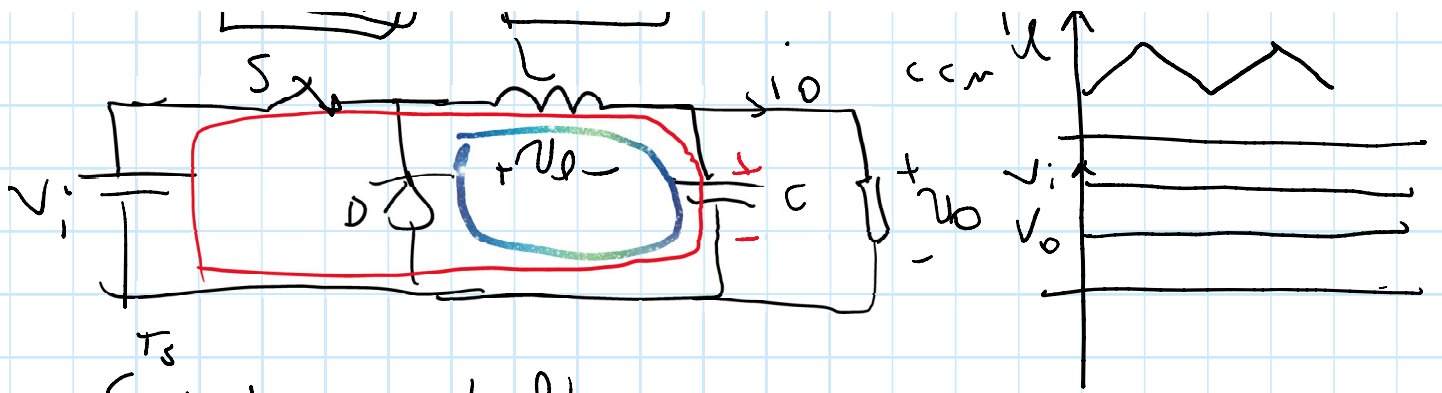
$$\bar{V}_o = \frac{T_{on} \times 0 + T_{off} \frac{V_i}{1-D}}{T_{on} + T_{off}} = \bar{V}_i \Rightarrow \boxed{M=1}$$

switching power supplies

buck / boost / buck-boost

buck converter $|V_o| < V_i$ pulse





$$\oint N_x dz = \text{net flux}$$

$$\int_{t_{on}}^{t_{on}+T_s} N_L dz + \int_{t_{on}+T_s}^{T_s} N_L dz = \text{Net flux} = 0 \quad \text{steady state}$$

$$\int_{t_{on}}^{t_{on}+T_s} (V_i - V_o) dz + \int_{t_{on}+T_s}^{T_s} (-V_o) dz = 0$$

f_s is high
 C is large
 V_o is constant

$$(V_i - V_o)t_{on} + (-V_o)t_{off} = 0$$

$$\Rightarrow V_o = \frac{t_{on}}{T} V_i = D_s V_i$$

Design Criteria: MOSFET / IGBT S.P. 1.5

$$V_{CE(sus)} \geq V_{i,max} \propto S.F.$$

$$I_C(A) \geq$$

$$V_{CE} \geq 150 \times 1.5 = 225V$$

$$\left\{ \begin{array}{l} V_i = 100 - 150V \\ P_o = 100W - 500W \\ \Delta V_o < 5\% \\ \eta, \text{ cost.} \\ f_s, V_{oc} = 60V \end{array} \right.$$

$$\text{lossless system: } P_{in} = P_o = V_o I_o$$

$$P_{in} = P_o = V_i I_i = V_i \times I_c \Rightarrow I_c = \frac{P_{in} = P_o}{V_i}$$

$$I_{in} \approx I_o \approx V_i I_i \approx V_i \times I_c \Rightarrow I_c = \frac{V_i I_o}{V_i}$$

$$I_{(AV)} \geq \frac{P_{max}}{V_{o min}} = \frac{500W}{100} = 5A \quad \text{S.P.} = 7.5$$

$D \approx ?$ fast recovery

$$V_{ORM} \geq V_{i max} \times S.P. = 225V$$

$$I_{(AV)} \geq$$

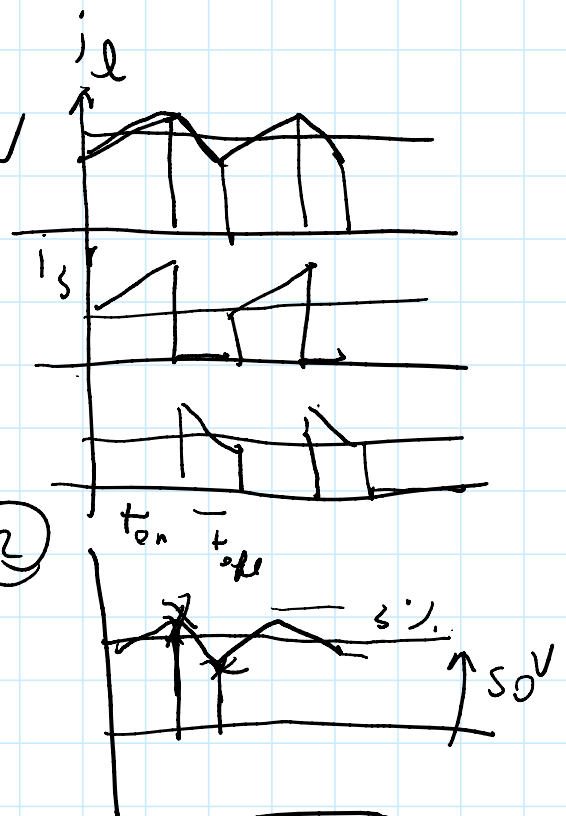
$$\bar{I}_L, \bar{I}_S, \bar{I}_D$$

$$\bar{I}_S = D_S \bar{I}_L \quad \textcircled{3} \quad \bar{I}_D = (1 - D_S) \bar{I}_L \quad \textcircled{2}$$

$$D_{S min} < D_S \leq D_{S max} \quad N_i = 100 - 150$$

$$\frac{50}{150} \leq D_S \leq \frac{50}{100}$$

$\textcircled{1}$



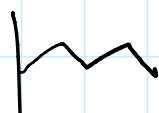
$$\underline{C}: V_o = 50V, \Delta V_o (\%) = 2.5\%$$

$$\frac{1}{2} C V_1^2 - \frac{1}{2} C V_2^2 = P_o \times t_{eff}$$

$$\frac{1}{2} C [(50 + 1.25)^2 - (50 - 1.25)^2] = P_{max} \times t_{eff max}$$

CV ✓

$$\boxed{L = 9.}$$



$$\boxed{L, f_s, \Delta i_L} \quad \text{per CCM}$$



